

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 16: Water: A Precious Resource

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	16 — Water: A Precious Resource
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Every distractor is a plausible near-miss — surface coverage read as usable supply, recharge rate confused with stored volume, falling-table symptoms mistaken for causes, balance maths inverted, drip/harvesting effects reversed, and the cycle's constant total misread as creation or loss. Do not write on this paper — mark answers on your answer sheet.

1. A sealed glass jar holds a little water at the bottom; over a hot day droplets appear on the inside of the lid, and by a cold night a thin frost coats the lid. The water inside the jar has, in turn, gone through which set of processes?

- (A) condensation, then freezing, then evaporation, in that order
- (B) freezing first, then melting, then evaporation into the lid
- (C) evaporation from the bottom, then condensation on the lid, then freezing of those droplets
- (D) transpiration, then infiltration, then precipitation inside the jar

2. A child insists that the steam seen above a kettle and the invisible moisture that later fogs a cold window are two different things. The accurate view is that:

- (A) both are water — the visible cloud is condensing droplets while the invisible moisture is true vapour, two appearances of the same substance changing state
- (B) the visible steam is water but the invisible moisture is air, a different substance
- (C) the invisible moisture is water but visible steam is a new gas made by heating
- (D) they are genuinely two separate chemicals that only behave alike near a kettle

- 3.** Ice floating in a pond, the liquid water beneath it, and the moisture in the air just above all coexist on the same winter day. This best illustrates that water:
- (A) must be in only one state across an entire region at any moment
 - (B) can exist in all three of its states at the same time when conditions differ from place to place
 - (C) changes into a different substance each time it changes state
 - (D) exists as ice only at the poles and never beside liquid water
- 4.** Across a full year, scientists find the total quantity of water on Earth barely changes even though oceans evaporate, clouds rain down, and rivers run to the sea every single day. The reason is that the water cycle:
- (A) creates new water through rain at about the same rate that the Sun destroys it
 - (B) loses water to space each day but gains an equal amount from the Sun
 - (C) freezes exactly as much new water as melts, so nothing else matters
 - (D) keeps moving the same water around between sea, air and land, changing its form and place but not its overall amount
- 5.** On a humid evening the air over a lake is thick with vapour, yet no cloud forms until the air rises and cools higher up. The single change that turns that vapour into a cloud is:
- (A) further heating of the vapour so it expands into a visible cloud
 - (B) the vapour freezing directly on the lake surface
 - (C) cooling of the rising vapour until it condenses into countless tiny droplets
 - (D) the vapour being pulled down and infiltrating the soil
- 6.** A dense forest releases far more moisture into the air over a day than a bare rocky hill of the same size. The chief reason is that the forest adds the extra moisture through:
- (A) transpiration — water drawn up by many trees and released as vapour from their leaves
 - (B) condensation, because leaves are cooler than bare rock
 - (C) infiltration, because forest soil pushes water upward into the air
 - (D) precipitation, because forests make their own private rainfall
- 7.** Place these four events in the order they occur for a single parcel of ocean water completing one loop of the water cycle: it falls on land as rain; it rises as vapour; it gathers into a cloud; it flows back to the sea in a river.
- (A) gathers into a cloud, rises as vapour, flows to the sea, falls as rain
 - (B) falls as rain, rises as vapour, flows to the sea, gathers into a cloud
 - (C) rises as vapour, gathers into a cloud, falls as rain, flows back to the sea
 - (D) flows to the sea, falls as rain, gathers into a cloud, rises as vapour
- 8.** Hail, drizzle and a snowfall look very different, yet a teacher groups all three under one heading of the water cycle. The shared reason is that each is a way that:

- (A) water leaves the ground for the air, so all three count as evaporation
- (B) water returns from the air to the ground, so all three count as precipitation
- (C) vapour is forming high up, so all three count as condensation
- (D) plants are releasing moisture, so all three count as transpiration

9. A poster reads: 'Water covers about 71% of Earth's surface — so we have water to spare.'
A careful student rewrites it correctly as:

- (A) since 71% of the surface is water, about 71% of it is fresh and ready to drink
- (B) the 71% figure proves Earth has almost no water and we are nearly dry
- (C) most of that surface water is salty ocean, leaving only a tiny fresh, usable fraction, so we cannot take spare water for granted
- (D) surface coverage equals usable supply, so 71% of our water is drinkable

10. If every drop of Earth's water were poured into 1000 identical jugs, the number of jugs holding salty ocean water would be closest to:

- (A) about 710 jugs
- (B) about 500 jugs
- (C) about 30 jugs
- (D) about 970 jugs

11. A guide claims, 'Glaciers and ice-caps store most of Earth's fresh water, so they are where our drinking water mainly comes from each day.' The correct refinement is:

- (A) glaciers hold most fresh water and we indeed drink directly from glacier ice daily
- (B) glaciers do hold most fresh water, but it is locked as ice, so day-to-day supply comes mostly from groundwater and surface sources
- (C) glaciers hold almost no fresh water, so the claim is wrong on both counts
- (D) all our daily drinking water is taken straight from the salty oceans

12. Rank these three from the largest store of water to the smallest, using chapter knowledge: water frozen in ice-caps and glaciers, water in the oceans, water flowing in all the world's rivers right now.

- (A) ice-caps and glaciers, then oceans, then river water
- (B) river water, then oceans, then ice-caps and glaciers
- (C) oceans, then ice-caps and glaciers, then river water
- (D) oceans, then river water, then ice-caps and glaciers

13. Two districts in the same state get equal rainfall, yet one suffers shortage while the other has plenty. The best lesson is that fresh water is:

- (A) always equal wherever rainfall is equal, so one report must be mistaken
- (B) found only in the district with the larger population
- (C) spread perfectly evenly across Earth at all times
- (D) distributed unevenly, depending also on how much is stored, drawn out and recharged in each place

14. A factsheet states India holds about 17% of the world's people but only roughly 4% of its fresh water. Read together, the two figures most directly mean that, on average, an Indian shares from a:

- (A) far larger slice of fresh water than the world average, signalling surplus
- (B) much smaller slice of fresh water than the world average, signalling real water stress
- (C) slice exactly equal to the world average, so there is no special pressure
- (D) slice that grows automatically as the population grows

15. A driller's log notes: '0 to 9 m, pores held air; below 9 m every pore was full of water.' The 9 m mark is best named the:

- (A) aquifer — the porous rock that yields the water
- (B) water table — the depth where the pore spaces first become completely water-filled
- (C) infiltration depth — where rain stops moving downward
- (D) spring line — where water first leaves the ground at the surface

16. Three nearby wells are bored into three different layers: thick porous sand that gushes water, solid granite that stays dry, and clay that yields only a trickle. Which layer is acting most fully as an aquifer?

- (A) the solid granite, because hard rock holds the most water
- (B) the clay, because it traps water and never lets it move
- (C) all three equally, since any underground layer is an aquifer
- (D) the porous sand, because it both stores groundwater and lets it move freely to the well

17. Rain falls steadily on bare, open soil for many hours with little running off. The single process that carries this rain down to join the groundwater is:

- (A) evaporation back up from the wet soil surface
- (B) transpiration through the leaves of nearby plants
- (C) infiltration — the slow seeping of water down through the soil pores
- (D) condensation of vapour onto the cool soil grains

18. A natural spring flows from a hillside all year, while a hand-dug well a short distance lower runs dry every summer. The most consistent explanation is that:

- (A) the spring makes its own water, so it never depends on the water table
- (B) the well is deeper than the spring, which is why it dries out first
- (C) the spring taps groundwater that stays above its outlet, while the well's bottom sits above the summer water table
- (D) summer rain reaches the spring but skips the well entirely

19. In a flat plain, neighbours notice that wherever the land dips into a low pond the water table is reached at shallow depth, while on the higher mounds wells must be dug far deeper. This pattern shows that the water table:

- (A) is perfectly flat everywhere underground regardless of the surface shape
- (B) is always deepest exactly under the lowest pond
- (C) rises only under the tallest mounds and nowhere else
- (D) roughly follows the shape of the land, sitting nearer the surface in low spots than under high ground

20. After installing a powerful tube well, a farm draws water from far deeper than the old open dug well ever could. By itself, switching to a deeper tube well does which of the following to the area's water table?

- (A) nothing directly — it reaches deeper water but does not raise the table; heavy pumping can in fact pull it down
- (B) raises the water table, because deeper wells push water upward
- (C) instantly refills the aquifer by reaching new water
- (D) makes the groundwater fresh that was previously salty

21. An aquifer beneath a town gains 60 units of water each year by recharge and loses 60 units to pumping. The town then doubles its pumping while recharge stays the same. Over the following years the water table will:

- (A) fall, because each year far more is now drawn out than seeps back in
- (B) stay level, since doubling pumping also doubles the recharge
- (C) rise, because faster pumping draws more rain into the ground
- (D) fall one year then settle back on its own to the old level

22. A village log shows yearly recharge of 40 units and withdrawal of 25 units for several seasons. The expected effect on the water table is that it:

- (A) slowly falls, because any pumping always lowers the table
- (B) stays exactly level, since some water is always pumped
- (C) slowly rises, because more water enters the aquifer each year than is taken out
- (D) rises one year then must fall the next to balance

23. Engineers want a city's water table to hold steady year after year. The condition that must be met over the long run is that:

- (A) withdrawal must always exceed recharge by a small, safe margin
- (B) the water drawn out each year is roughly matched by the water recharged in
- (C) recharge must be kept at zero so the stored water is never disturbed
- (D) rainfall must be prevented so the aquifer cannot leak away

24. Town P recharges 70 and withdraws 90 units a year; Town Q recharges 90 and withdraws 70. Same starting table, same rainfall. After several years, which prediction holds?

- (A) P's table falls while Q's rises, because P runs a yearly deficit and Q a surplus
- (B) both tables stay identical, since their numbers are mirror images

- (C) Q's table falls faster, because larger recharge means more disturbance
- (D) P's table rises, because pumping more brings up more water

25. A region's recharge has been steady, but each year a few more bore wells are added and each pumps the same as the rest. The yearly trend in the gap between water taken out and water put back is that the gap:

- (A) narrows, so the water table slowly recovers
- (B) stays constant, leaving the water table unchanged
- (C) widens, pushing the water table progressively lower
- (D) vanishes, because more wells force more recharge

26. Over four years a monitored aquifer's stored volume reads 100, 92, 80, 64 units while yearly recharge was unchanged. The most consistent explanation is that withdrawal each year was:

- (A) greater than recharge, and the yearly gap kept growing, so the store fell ever faster
- (B) less than recharge, so the store should have risen instead
- (C) exactly equal to recharge, keeping the store level
- (D) zero, since nothing was being pumped out at all

27. A growing city sees its population, its factories and its paved area all rise sharply over a decade, while rainfall is unchanged. The combined effect on the water table is that it:

- (A) falls, because demand and withdrawal rise while paving cuts the recharge
- (B) rises, because more people and roads attract more rain to soak in
- (C) stays level, because extra demand is always matched by extra rain
- (D) turns salty, although its depth does not change

28. A planner argues that paving every open lot will not hurt groundwater because 'rain still falls on the city the same as before.' The flaw is that paving:

- (A) actually increases how much rain soaks into the sealed ground
- (B) turns the falling rain salty before it can reach the soil
- (C) has no effect on recharge as long as the rainfall total is unchanged
- (D) stops the rain from soaking in, so it runs off into drains and no longer recharges the aquifer

29. Clearing a hillside forest is followed, a few seasons later, by drier wells in the valley below. The best causal chain is:

- (A) fewer trees cool the soil, the rain freezes, and the wells ice over
- (B) fewer trees stop rainfall completely, so no water reaches the valley
- (C) fewer trees make the rain salty, so the valley wells turn undrinkable
- (D) fewer trees hold back and slow the rain less, so more runs off and less soaks in, recharge drops and the valley water table falls

30. Of these four scenarios, which would drive a region's water table DOWN the fastest over a few years?

- (A) good rainfall on open soil with many new recharge pits
- (B) scanty rainfall together with a booming population, new factories and widespread concreting
- (C) drip irrigation on all farms plus rooftop harvesting everywhere
- (D) tree planting, leak repairs and keeping the ground unpaved

31. Two identical plots get the same rain. Plot M is left as open garden soil; Plot N is fully tiled and its rain is piped to a street drain. After several monsoons, the water table is higher under:

- (A) Plot N, because the tiles trap and store the rain underground
- (B) Plot M, because open soil lets rain infiltrate and recharge the groundwater
- (C) neither, because tiled and open ground recharge the same way
- (D) Plot N, because the drain carries water back into the aquifer

32. A factory keeps drawing the same groundwater each year, but the surrounding fields are gradually replaced by a paved logistics park. Even with steady pumping, the water table now falls faster than before because:

- (A) the factory must be secretly pumping more than it reports
- (B) the paving makes the groundwater salty, lowering its level
- (C) the paving cuts recharge, so the unchanged withdrawal now outpaces a smaller inflow
- (D) the rainfall over the park must have dropped at the same time

33. A village's open dug-well stayed full for thirty years but ran dry this summer, although last year's monsoon was normal. The most likely chapter-based cause is:

- (A) one normal monsoon permanently ruined the aquifer below the well
- (B) years of over-extraction in the area have lowered the water table below the well's bottom
- (C) the well water evaporated straight up through the surrounding rock
- (D) the well suddenly turned salty and so could no longer hold water

34. As a district's water table keeps sinking year after year, which set of consequences fits best?

- (A) shallow wells dry up, deeper wells and tube wells must be sunk, and pumping costs rise
- (B) wells overflow and pumping becomes effortless and free
- (C) the rainfall automatically increases each year to make up the loss
- (D) the soil above the aquifer slowly turns permanently to ice

35. A town's leaders see deeper wells, costlier pumping and dry shallow wells, and conclude these problems are the CAUSE of their water trouble. Their reasoning error is

that these are actually:

- (A) the true causes, so deepening the wells will solve everything
- (B) symptoms of an already-falling water table, not its cause
- (C) unrelated to the water table and merely a coincidence
- (D) signs that the water table is rising too quickly

36. Coastal wells that once gave fresh water turn salty after years of heavy pumping. Within the chapter's logic, the reason is that:

- (A) the act of pumping chemically converts fresh water into salt water
- (B) the coastal rain is naturally salty and refilled the wells with brine
- (C) lowering the fresh groundwater allowed nearby salty water to move in toward the wells
- (D) the concrete near the coast manufactures salt within the soil

37. A water-stressed town's audit shows its problems all trace back to drawing more groundwater than the rains replace. Which response treats the CAUSE rather than just the symptoms?

- (A) fitting every well with a more powerful pump to lift the deeper water
- (B) reducing over-extraction and boosting recharge so withdrawals no longer outrun inflow
- (C) deepening each well repeatedly to chase the dropping water down
- (D) trucking in bottled water for the town's daily drinking needs

38. A house leads its rooftop rain through a downpipe into a gravel-and-sand pit so the water slowly seeps underground. This pit raises the water table more than a closed drain would because:

- (A) it chemically transforms the rooftop rain into fresh groundwater
- (B) it pumps stored water upward out of the soil for later use
- (C) it heats the rain so the warmer water sinks down on its own
- (D) it lets the collected rain infiltrate into the aquifer instead of being carried away as runoff

39. Traditional structures such as the bawri (step-well), tanka and johad were built across dry parts of India chiefly to:

- (A) collect, store or recharge rain and groundwater for use through the dry season
- (B) generate electricity from the force of falling water
- (C) remove the salt from sea water for inland villages
- (D) measure the exact daily rainfall of each region

40. A village builds an earthen johad across a seasonal stream. The main way it lifts the water table is by:

- (A) pumping groundwater up to the surface for the village to use
- (B) holding back the rain water so it has time to seep into the ground and recharge the aquifer
- (C) sealing the streambed so that no water can ever soak through it
- (D) spreading the stored water out so it evaporates quickly into the air

41. A farmer switches a field from flooding to drip irrigation. Compared with before, the field now:

- (A) uses far less water by delivering it drop by drop to the roots, easing the demand on groundwater
- (B) uses much more water and so lowers the water table even faster
- (C) needs no irrigation water at all once the drip pipes are laid
- (D) makes the surrounding groundwater turn salty as a side effect

42. Besides cutting water use, drip irrigation tends to leave fewer weeds in the dry strips between crop rows because:

- (A) the drip water carries a built-in chemical weedkiller
- (B) the slow drops flood the rows and wash all the weeds away
- (C) drip irrigation freezes the bare soil so nothing can sprout
- (D) only each crop plant's root zone is kept moist, so weeds in the dry gaps get little water

43. A market gardener must water 200 potted plants on a hot day using as little water as possible. The best method is:

- (A) flooding the whole bench so the water reaches every pot
- (B) an overhead sprinkler left running across the pots all afternoon
- (C) hosing water generously over the leaves of all the plants
- (D) a drip line releasing water slowly right at each pot's roots

44. Why is rooftop rainwater harvesting promoted especially strongly in big cities?

- (A) city roofs and pavements shed huge volumes of rain that would otherwise run off without recharging the ground
- (B) cities by their nature receive far more rainfall than any village
- (C) city groundwater is endless, so harvesting is just a precaution
- (D) harvesting only works at all where there is no soil whatsoever

45. A planner wants to pick the single most reliable measure to slow a steadily falling water table. The best choice is:

- (A) sanctioning more bore wells so the demand is shared out
- (B) tiling over the city's open grounds to keep them clean
- (C) routing all storm water out of the city as quickly as possible
- (D) requiring rainwater-harvesting recharge structures on new buildings

46. A town's water-use survey reads: drinking and cooking 6%, washing and bathing 24%, irrigation of nearby farms 62%, industry 8%. To ease groundwater stress most, the town should first target:

- (A) drinking and cooking, since saving on the smallest use helps the most
- (B) industry, because 8% is somehow the largest share to cut
- (C) nothing, because none of these uses can be reduced at all
- (D) irrigation, since it is by far the largest use and can shift to water-saving drip methods

47. A school finds many taps each leaking a little, adding up to about 30 litres lost daily. The most sensible response is to:

- (A) ignore them, because a single small drip is far too little to matter
- (B) open every leaking tap fully so it can no longer drip
- (C) mend the leaks promptly, since many small drips add up to a large daily loss
- (D) leave a bucket under each tap to catch the drips forever

48. Which set of household actions ALL work to conserve fresh water?

- (A) using a bucket-and-mug bath, closing the tap while brushing, and mending leaking taps
- (B) taking long showers, leaving the tap running while soaping, and filling the bathtub
- (C) paving the garden, flood-irrigating the lawn, and leaving taps dripping
- (D) draining the rooftop rain to the road, over-watering pots, and running the hose

49. A reservoir reads full, then a little down, then lower, then lowest across four years, while rainfall stayed normal but the town drilled many new tube wells each year. The pattern best shows that:

- (A) recharge has outpaced withdrawal, so storage should be climbing
- (B) withdrawals have outpaced recharge, steadily depleting the stored water
- (C) rainfall must have failed completely in every one of the four years
- (D) the falling level proves the stored water has turned to vapour

50. Which observation would be the clearest sign that a recharge programme is actually WORKING in an area?

- (A) the nearby ocean gradually becomes noticeably less salty
- (B) the monsoon rainfall suddenly doubles compared with past years
- (C) wells that had gone dry begin giving water again at shallower depths over a few seasons
- (D) the local rivers stop flowing entirely during the dry season

51. Concreting over a floodplain that used to soak up the monsoon typically produces which twin effect?

- (A) less flooding during the rain and more recharge afterwards
- (B) more surface flooding during the rain and less groundwater recharge afterwards

- (C) no flooding and no change at all in the recharge
- (D) more recharge during rain but also far more flooding

52. A region installs rooftop harvesting (raising recharge) and converts its farms to drip (cutting withdrawal), so that recharge now clearly exceeds withdrawal. Over time the water table will:

- (A) fall, because two interventions together drain the aquifer faster
- (B) stay flat, since recharge and withdrawal are forced to always match
- (C) rise only for as long as the two measures are later removed
- (D) rise, because more water now enters the aquifer than leaves it

53. Farm A floods its fields and its well dries by April; neighbouring Farm B drips its fields, on the same soil and rainfall, and its well lasts until June. The fairest conclusion is that:

- (A) flood irrigation conserves more groundwater than drip does
- (B) the irrigation method has no bearing on how long a well lasts
- (C) Farm B's lower water use let its well outlast Farm A's
- (D) Farm B must simply have received much more rain than Farm A

54. A city official claims, 'If we double the number of tube wells, we double our water supply permanently.' The chapter-based flaw is that:

- (A) tube wells only withdraw from a limited aquifer, so pumping faster than it recharges soon lowers the table and dries the wells
- (B) adding tube wells creates brand-new water out of nothing
- (C) the aquifer is unlimited, so the doubling plan is perfectly safe
- (D) tube wells add rain to the sky to keep refilling themselves

55. A village runs four measures at once: rooftop harvesting, drip on its farms, repairing leaking mains, and planting trees on open ground. Over several years its water table most likely:

- (A) falls faster than before, because more structures mean more water use
- (B) rises, because recharge improves while wasteful withdrawals fall
- (C) turns salty as a combined side effect of the four measures
- (D) stays exactly the same forever, untouched by any of the measures

56. Two villages get identical rainfall yet end up with very different water tables after a decade. The best explanation is that they may differ in:

- (A) nothing — equal rainfall must force their water tables to be identical
- (B) which village has more concrete, since concrete always raises the table
- (C) how much they pump out and how much rain they let infiltrate versus run off
- (D) how deep they dig their wells, since deeper wells raise the table

57. World Water Day is observed each year to remind everyone to value and conserve water. It falls on:

- (A) 5 June
- (B) 22 April
- (C) 14 November
- (D) 22 March

58. An orchard that once kept its valley well brimming is cleared for a car park; within years the valley well struggles in summer. Beyond losing runoff control, the loss of the trees themselves matters because trees had been:

- (A) pumping groundwater upward to keep the well topped up
- (B) creating new water inside their roots for the aquifer
- (C) blocking the rain from ever reaching the soil at all
- (D) helping rain slow down and soak in, supporting the recharge that fed the well

59. During a long water shortage, a row of vegetable plants is left without enough water and the season's yield falls sharply. The chapter's explanation is that:

- (A) plants need water to grow and produce, so a shortage stunts their growth and lowers the crop
- (B) less water makes the plants grow faster and yield more than usual
- (C) the plants turn the missing water into extra vapour and thrive
- (D) a water shortage makes the plants saltier rather than smaller

60. Which statement best captures the chapter's central message that 'water is a precious resource'?

- (A) usable fresh water is genuinely scarce, so we must use it wisely and manage it carefully
- (B) water is effectively unlimited, so making an effort to save it is pointless
- (C) only farmers, not ordinary households, need to bother conserving water
- (D) the sea will always supply more than enough drinking water for everyone

Solutions — Science (Class 7), Chapter 16: Water: A Precious Resource — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	A	31	B	41	A	51	B
2	A	12	C	22	C	32	C	42	D	52	D
3	B	13	D	23	B	33	B	43	D	53	C
4	D	14	B	24	A	34	A	44	A	54	A
5	C	15	B	25	C	35	B	45	D	55	B
6	A	16	D	26	A	36	C	46	D	56	C
7	C	17	C	27	A	37	B	47	C	57	D
8	B	18	C	28	D	38	D	48	A	58	D
9	C	19	D	29	D	39	A	49	B	59	A
10	D	20	A	30	B	40	B	50	C	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) The water first turns to vapour rising from the warm base (evaporation), the vapour cools and forms droplets on the lid (condensation), and the cold night turns those droplets to solid ice (freezing). The sequence must begin with vapour leaving the liquid; transpiration and infiltration belong to plants and soil, not a sealed jar.

2. (A) Both are the single substance water. True water vapour is invisible; what we call steam above a kettle is already tiny condensed droplets. Neither is a different substance or a newly created gas — water simply shifts between states without becoming something else.

3. (B) Different micro-conditions let solid, liquid and vapour coexist together. Water is not restricted to one state region-wide, it does not become a different substance when it changes state, and ice can plainly sit on liquid water far from the poles.

4. (D) The cycle recirculates a fixed pool of water; only its form and location change while the total stays nearly constant. Water is neither created by rain, destroyed by the Sun, nor

lost to space and topped up — those are misreadings of a closed cycle.

5. (C) A cloud appears when vapour is cooled enough to condense into droplets — cooling, not heating, is the trigger. Freezing on the surface and infiltration into soil are unrelated to cloud formation aloft.

6. (A) Trees pull up water and let it escape as vapour from their leaves — transpiration — so a forest adds far more moisture than bare rock. Condensation removes vapour rather than adding it, and infiltration and precipitation do not push water up from leaves.

7. (C) The parcel must first leave the sea as vapour, then condense into a cloud, then fall as rain on land, then return seaward in a river. Any order that puts cloud or rain before evaporation breaks the physical sequence.

8. (B) Hail, drizzle and snow are all forms of water falling from the air to the surface, which is precipitation. They are not evaporation (ground to air), nor condensation (cloud formation), nor transpiration (release from leaves).

9. (C) Surface coverage is not the same as usable supply. Roughly 97% of all water is salty ocean and most of the rest is frozen or deep, so the fresh usable share is tiny. Coverage cannot be read as either a fresh-water percentage or proof of scarcity of water overall.

10. (D) Since roughly 97% of all water is in the oceans, about 970 of the 1000 jugs would be salty. The 71% figure is surface coverage, not a share of the water itself, and the small remainder is the fresh fraction, not the ocean share.

11. (B) Glaciers store the largest share of fresh water, but it is frozen and not readily tapped, so everyday use draws on groundwater and surface water. The claim is only half right — the storage fact is correct, the daily-source part is not — and oceans are salty, not a drinking source.

12. (C) Oceans hold by far the most water; the frozen ice-caps and glaciers are the largest fresh store next; the water in rivers at any instant is the smallest of the three. Putting ice or rivers ahead of the oceans, or rivers ahead of glaciers, reverses the true scale.

13. (D) Equal rain does not guarantee equal availability, because storage, withdrawal and recharge differ between places — fresh water is unevenly distributed. It is not fixed by rainfall alone, not tied to population size, and certainly not spread evenly everywhere.

14. (B) A large share of people drawing on a much smaller share of water means each person has less than the world average — clear water stress. The numbers cannot mean surplus, an equal share, or a slice that conveniently expands with population.

15. (B) The upper level at which the underground pores are fully saturated is the water table. The aquifer is the water-bearing rock layer itself, infiltration is the seeping process,

and a spring is where groundwater emerges at the surface — none names that saturation level.

16. (D) An aquifer is a porous layer that stores water and readily gives it up, which fits the gushing sand. Solid granite holds little and yields little, clay traps water without releasing it, and not every underground layer qualifies as an aquifer.

17. (C) Water seeping downward through the soil to reach groundwater is infiltration. Evaporation and transpiration move water upward into the air, and condensation forms droplets rather than carrying rain underground.

18. (C) A perennial spring keeps flowing because the water table stays above its mouth, whereas a well goes dry when the table drops below its base. Springs do not manufacture water, the failing well is the shallower one, and seasonal rain does not selectively skip a single well.

19. (D) The water table generally mirrors the surface, lying closer to the top in hollows and deeper beneath rises. It is not a uniform flat sheet, it is not deepest under the lowest ground, and it does not rise selectively under high mounds.

20. (A) A deeper well only reaches lower water; it does not lift the table, and pumping hard from it tends to lower the table further. It cannot push water up, create new aquifer water, or change salty water to fresh.

21. (A) With withdrawal now at 120 against recharge of 60, the deficit drains stored water and the table falls year on year. Pumping does not double recharge, does not pull extra rain underground, and the imbalance does not self-correct.

22. (C) When recharge of 40 exceeds withdrawal of 25, stored water builds up and the table climbs. The presence of some pumping does not force a fall, and a surplus does not have to reverse itself to 'balance' — it simply accumulates.

23. (B) A steady table needs long-run balance: outflow about equal to inflow. Letting withdrawal exceed recharge would steadily deplete it, zero recharge would starve it, and blocking rainfall removes the very source of recharge.

24. (A) P loses 20 units a year and Q gains 20, so P's table drops and Q's climbs despite the mirrored figures. Mirror-image numbers do not cancel between separate towns, larger recharge raises rather than lowers a table, and over-pumping lowers rather than raises P's.

25. (C) Adding wells while recharge is fixed increases withdrawal, so the deficit grows and the table sinks faster each year. More wells do not narrow the gap, keep it constant, or somehow create extra recharge.

26. (A) Falling stores that drop by larger amounts each year (8, then 12, then 16) mean withdrawal exceeded recharge by a widening margin. Equal or smaller withdrawal would have held or raised the store, and zero withdrawal could not drain it.

27. (A) More people and industry pump more water out while paving reduces how much rain seeps in, so withdrawal climbs as recharge drops and the table falls. Roads do not attract rain underground, demand is not auto-matched by rainfall, and the issue here is depth, not salinity.

28. (D) What matters for recharge is how much rain can infiltrate, and paving sends it running off instead of into the ground. Sealed surfaces reduce, not increase, infiltration; they do not salt the rain; and recharge depends on infiltration, not merely on how much rain falls.

29. (D) Without trees to slow and hold the rain, more water runs off and less infiltrates, cutting recharge and lowering the valley table. Deforestation does not freeze soil, halt rainfall, or salt the rain.

30. (B) Little inflow with soaring withdrawal and almost no infiltration is the worst case and drains the table quickest. The other three combine more recharge with less waste, which protects or raises the table.

31. (B) Open soil lets rain soak in and recharge the table, while tiling plus a drain ships the water away as runoff. Tiles do not store rain underground, the two surfaces do not recharge equally, and a street drain removes water rather than returning it to the aquifer.

32. (C) When recharge shrinks but withdrawal stays the same, the balance tips and the table drops faster — the paving, not extra pumping or salinity or assumed less rain, is the cause.

33. (B) A long, gradual fall from sustained over-pumping eventually drops the table beneath the well, drying it despite a normal monsoon. A single normal monsoon does not wreck an aquifer, water does not evaporate up through rock, and salinity would not make a well refuse to hold water.

34. (A) A falling table forces wells deeper and makes lifting water harder and costlier while shallow wells fail. It does not cause wells to overflow, rainfall to self-adjust, or soil to freeze.

35. (B) Dry wells and pricier pumping are signs that the table has dropped — effects, not the root cause. Treating the symptoms by digging deeper ignores the real driver, they are not coincidental, and they certainly do not mean a rising table.

36. (C) Over-pumping drops the fresh water level so salty water encroaches into the same ground — the chapter's account of coastal over-extraction. Pumping does not change

water's chemistry, rain is not salty, and concrete does not make salt.

37. (B) The cause is withdrawal exceeding recharge, so cutting extraction and raising recharge is the real fix. Bigger pumps, deeper wells and imported water only cope with the falling table without addressing why it falls.

38. (D) A recharge pit holds the rain so it soaks down into the aquifer, while a drain sends it away. The pit does not change water chemically, it does not pump anything up, and warming is not what drives the seepage.

39. (A) These structures gather and store rain or help recharge groundwater so water lasts into dry months. They were not power plants, desalination works, or rain gauges.

40. (B) A johad ponds the runoff so it lingers and infiltrates, recharging the aquifer. It does not pump water up, seal off seepage, or speed evaporation — those would not raise the table.

41. (A) Drip delivers water straight to the roots with little loss, cutting use and easing groundwater demand. It does not use more water, eliminate watering altogether, or salinise groundwater.

42. (D) Drip wets only the crop's roots, leaving the gaps dry where weeds cannot thrive. There is no weedkiller in the water, drip does not flood, and it does not freeze the soil.

43. (D) Drip targets each pot's roots with minimal loss, the most water-thrifty choice. Flooding, all-day sprinkling and overhead hosing all spill and evaporate far more water.

44. (A) Hard urban surfaces send vast runoff to drains, so capturing it for recharge matters most in cities. Cities do not inherently get more rain, their groundwater is not limitless, and harvesting is not restricted to soil-free sites.

45. (D) Mandating recharge structures directly returns rain to the aquifer, slowing the decline. More bore wells increase withdrawal, tiling cuts recharge, and rushing storm water away wastes the rain that could have recharged the ground.

46. (D) Irrigation at 62% dominates the demand, so switching it to efficient drip yields the biggest saving. Drinking is the smallest slice, industry at 8% is not the largest, and uses can certainly be reduced.

47. (C) Small leaks combine into a sizeable daily waste, so fixing them quickly is the right move. Dismissing the drips ignores the total, opening taps fully wastes far more, and catching drips in buckets only stores the loss rather than stopping it.

48. (A) Bucket baths, shutting taps and mending leaks each cut waste, so all three conserve water. The other sets each list habits that increase use or runoff rather than save water.

49. (B) A steady decline despite normal rain points to extra tube wells drawing more than the rains replace, depleting storage. Recharge has not outpaced withdrawal, rainfall did not fail (it was normal), and water did not vapourise away.

50. (C) A recovering, shallower water table that revives dead wells is the direct evidence of successful recharge. Ocean salinity, rainfall amount and river flow are not measures of what a recharge programme does to groundwater.

51. (B) Sealing the floodplain blocks infiltration, so water that once soaked in now pools as floods and the aquifer gains less. It cannot both flood more and recharge more, nor reduce flooding while paved, nor leave recharge unchanged.

52. (D) With inflow exceeding outflow, stored water grows and the table climbs. Two water-saving measures do not drain an aquifer, recharge and withdrawal are not forced to match, and removing the measures would end the gain, not cause it.

53. (C) With soil and rainfall held equal, the only difference is method, so drip's lower use explains B's longer-lasting well. Flooding does not conserve more, method clearly does matter, and equal rainfall rules out the 'more rain' explanation.

54. (A) More tube wells just draw harder on a finite, slowly-recharged aquifer, so faster pumping depletes it and wells fail. Wells make no new water, the aquifer is not unlimited, and tube wells do not produce rain.

55. (B) Together these add recharge and cut waste, so inflow gains on outflow and the table rises. The measures save rather than spend water, they do not salinise it, and they certainly do change the table.

56. (C) Equal rain can give unequal tables when pumping and infiltration differ between the villages. Equal rain does not force equal tables, concrete lowers rather than raises a table, and well depth does not change the table's level.

57. (D) World Water Day is marked on 22 March. 5 June and 22 April are days set aside for the wider environment and the Earth, and 14 November is unrelated to water.

58. (D) Trees slow and hold rain so more infiltrates, aiding the recharge that sustained the well; removing them cuts that recharge. Trees do not pump water up, manufacture water, or stop rain from reaching the ground.

59. (A) Plants depend on water for growth and yield, so a shortage makes them wilt and produce less. Scarcity does not speed growth, plants cannot conjure vapour to thrive, and the result is stunting, not salinity.

60. (A) The chapter's core point is that fresh usable water is limited and must be managed with care. It is not unlimited, conservation is everyone's job not just farmers', and the salty

sea is not a drinking supply.